of the proper subset, the given statement is False.

13) Determine whether each of these statements are true or false

a) $x \in \{x\}$. The member of the set $\{x\}$ is x only. $\therefore$ The given statement is True.

b) $\{x\} \subseteq \{x\}$. By Theorem 1, the given statement is True.

c) $\{x\} \in \{x\}$. From part (a), we know that the given statement is False.

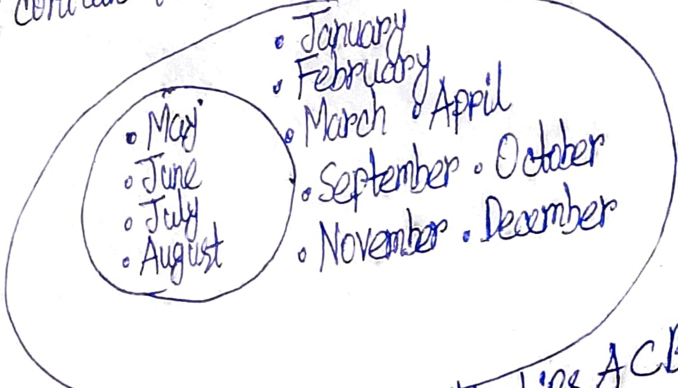
d) $\{x\} \in \{\{x\}\}$. The member of the set $\{\{x\}\}$ is $\{x\}$ only. $\therefore$ The given statement is True.

e) $\phi \subseteq \{x\}$. By Theorem 1, the given statement is True.

f) $\phi \in \{x\}$. From part (a), we know that the given statement is False.

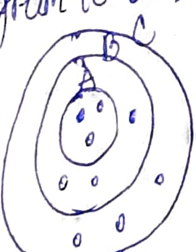
15) Use a Venn diagram to illustrate the set of all months of the year whose names do not contain the letter R in the set of all months of the year.

Solution:



17) Use a Venn diagram to illustrate the relationships $A \subset B$ and $B \subset C$.

Solution:



19) Suppose that A, B and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.

Solution: By Defn. 3, $\because A \subseteq B$ $\therefore \forall x(x \in A \rightarrow x \in B)$. By Defn. 3, $\because B \subseteq C$, $\therefore \forall x(x \in B \rightarrow x \in C)$. Let c be any arbitrary element and $c \in A$.

Steps	Reason
1. $\forall x(x \in A \rightarrow x \in B)$	Given
2. $\forall x(x \in B \rightarrow x \in C)$	Given